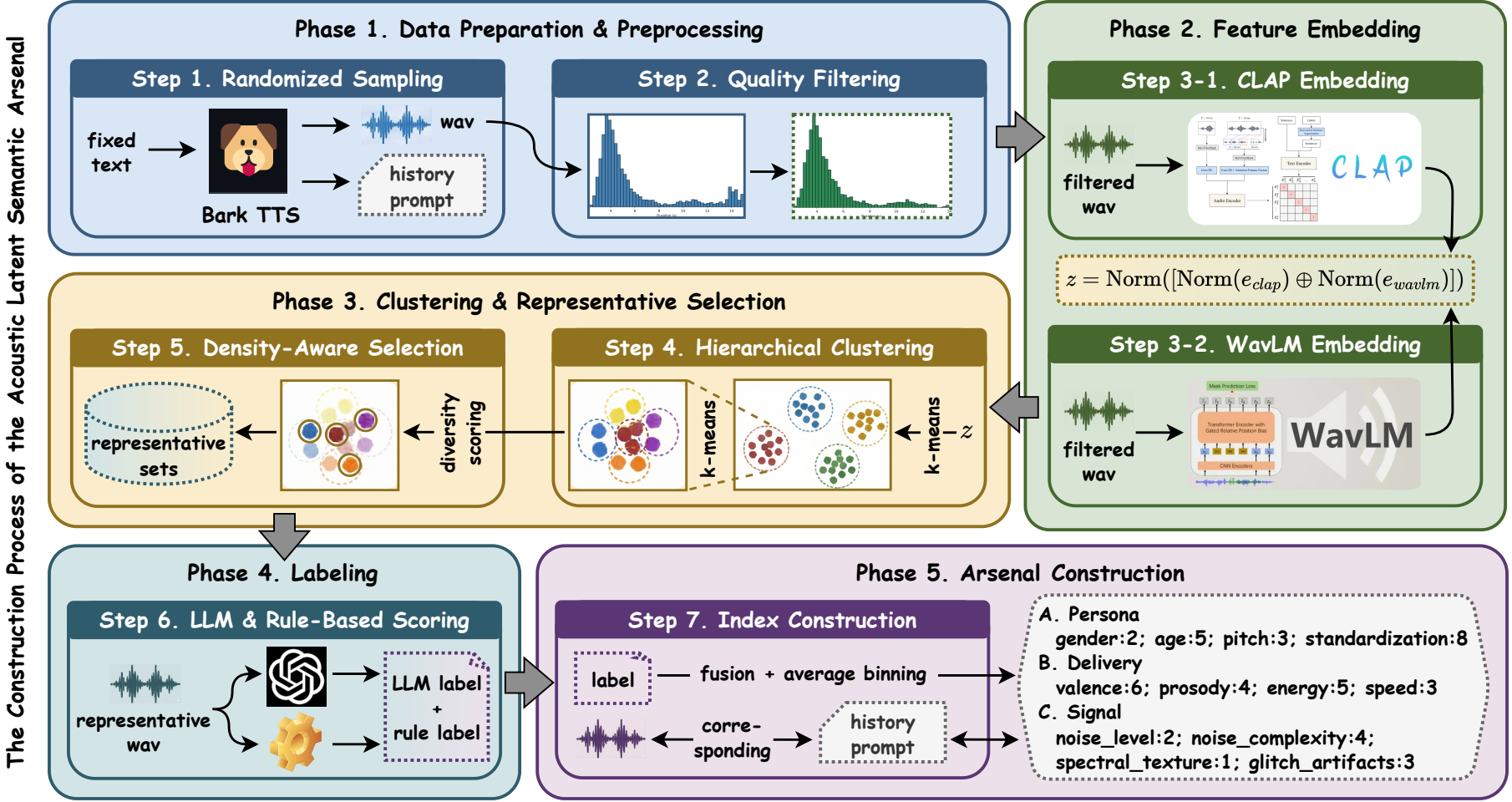
The Construction Process of the Acoustic Latent Semantic Arsenal
Phase 1. Data Preparation & Preprocessing
Step 1. Randomized Sampling
fixed text
Bark TTS
wav
history prompt
Step 2. Quality Filtering
Phase 2. Feature Embedding
Step 3-1. CLAP Embedding
filtered wav
CLAP
$z = \mathrm{Norm}([\mathrm{Norm}(e_{clap}) \oplus \mathrm{Norm}(e_{wavlm})])$
Step 3-2. WavLM Embedding
filtered wav
WavLM
Phase 3. Clustering & Representative Selection
Step 4. Hierarchical Clustering
k-means
k-means
z
Step 5. Density-Aware Selection
diversity scoring
representative sets
Phase 4. Labeling
Step 6. LLM & Rule-Based Scoring
representative wav
LLM label + rule label
Phase 5. Arsenal Construction
Step 7. Index Construction
label
fusion + average binning
corre-sponding
history prompt
A. Persona
gender:2; age:5; pitch:3; standardization:8
B. Delivery
valence:6; prosody:4; energy:5; speed:3
C. Signal
noise_level:2; noise_complexity:4;
spectral_texture:1; glitch_artifacts:3